

Inclusive AI Fashion Stylist with Measurement and Color-Blind Safety Analysis

Project ID: R26-ISE-001

Project Proposal Report

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MARCH 2026

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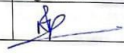
DECLARATION

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

This phenomenon is known as Color Vision Deficiency (CVD), commonly referred to as color blindness. CVD is found in a considerable percentage of the world's population and causes difficulties for people in distinguishing certain colors and shades of colors. For example, people with CVD face difficulties in distinguishing between red and green colors and between blue and yellow colors. This deficiency in distinguishing colors causes difficulties in making decisions in day-to-day life. For instance, people with CVD face difficulties in selecting clothes in the proper colors. Therefore, it is difficult for people with CVD to select proper colors without any assistance from others. Although various research works have been done in detecting color blindness and recognizing the colors of clothes, most of the existing solutions address these two issues independently. Currently, in various fashion and e-commerce websites, only objective color recognition is used to identify colors of clothes. Objective color recognition is based on the use of RGB values of colors. These existing solutions do not consider how people with different types of CVD perceive colors. To overcome this limitation, this research proposes the creation of a perception-aware clothing color identification system that is intended to assist individuals with Color Vision Deficiency. To do this, it first identifies what type of CVD the user has through a digital version of the Ishihara test. It pre-processes images of clothing uploaded by users using OpenCV to improve image quality before analysis. It employs a Convolutional Neural Network to identify the actual color of clothing. It also employs MobileNetV2 to improve the accuracy and efficiency of image classification. To make it perception-aware, a perception-aware dataset is first generated using the Colorspacious color vision simulation library to simulate CVD perception. The simulated images are then used to train a CNN model with MobileNetV2 transfer learning, enabling the system to learn the mapping between true garment colors and perceived colors under different CVD conditions. The contribution of this research is an intelligent system that displays clothing colors to individuals with CVD in a perception-aware manner to enable them to make more confident decisions when selecting clothing. From a social impact perspective, it promotes digital accessibility and inclusive design of technology products. From a commercial impact perspective, it has the potential to be applied to e-commerce sites to provide color blind individuals with accessible clothing color identification technologies.

Keywords: Color Vision Deficiency, Deep Learning, Clothing Color Detection, Perception-Aware Computing, Assistive Fashion Technology.

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to all those who supported me in completing this research proposal titled “Perception-Aware Clothing Color Identification and Decision Support for Individuals with Color Vision Deficiency.”

First and foremost, I extend my heartfelt appreciation to my academic supervisor and lecturers for their continuous guidance, constructive feedback, and valuable insights, which significantly strengthened the technical and research aspects of this study.

I am also thankful to my colleagues for their helpful discussions and suggestions during the planning phase of this project.

Finally, I would like to acknowledge the individuals who shared their experiences related to color vision deficiency. Their perspectives reinforced the importance of this research and helped validate the problem statement.

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LIST OF ABBREVIATIONS

CVD-Color Vision Deficiency

LMS – Long, Medium, Short (cone model in color perception)

OpenCV – Open-Source Computer Vision

PIL – Python Imaging Library

1. INTRODUCTION

1.1 Background and Context

Color plays an important role in fashion because it is related to self-confidence, personal identity, and acceptance by others. However, with the emergence of digital fashion platforms and shopping systems, consumers tend to focus on visual images when making fashion decisions. However, most fashion recommendation systems and online shopping systems have been developed based on the assumption that consumers perceive colors in the same way. This assumption creates accessibility problems for consumers with Color Vision Deficiency because they face difficulties in distinguishing colors due to abnormalities in the human retina's cone cells.

Color Vision Deficiency (CVD) primarily takes the form of red-green color deficiency, which includes protanopia and deutanopia, whereas tritanopia, or blue-yellow deficiency, is relatively rarer. These color perception limitations may lead to incorrect color identifications, which are likely to be very close in color, such as navy blue and black, or brown and green, making it challenging for color-blind people to correctly identify clothing colors and combinations of clothing that are likely to go well together. These issues are likely to be particularly challenging in online shopping scenarios, where color perception is solely dependent on images of clothes and not physically assessed. Therefore, color-blind people are likely to make incorrect choices in their clothing, leading to suboptimal purchasing decisions, which may result in monetary penalties in terms of return fees, embarrassment, and lack of self-confidence. Moreover, color-blind people may be forced to depend on others to confirm their color choices, thus limiting their personal freedom of expression.

Color Vision Deficiency is a very common condition that affects millions of people around the world. Research has suggested that 1 out of every 12 males and 1 out of every 200 females suffer from CVD. This makes the total number of people suffering from CVD around the world more than 300 million [1][11]. Out of the inherited CVDs, 98–99% involve red and green deficiency. This has a significant impact on color-based activities like dressing [12]. In South Asian countries like India and Sri Lanka, the prevalence of CVD is reported to be between 4% and 8% in males. Research has been conducted in Sri Lanka to study the prevalence of CVD in male school children, and the results have reported a prevalence of 3.1% [13]. Despite the prevalence of CVD in the population, there is little awareness about the impact of CVD on our day-to-day activities like fashion and online shopping.

Existing literature on CVD detection primarily deals with three different areas of CVD detection, color detection and color perception modelling. Digital Ishihara-based systems are commonly used for detecting CVD type but are primarily used for detection and diagnosis purposes rather than providing any aid in real-world decision-making scenarios [2][3][4]. Computer vision systems are also available for detecting the dominant color of garments from images but are based on the color perception of all humans [5][10]. Color perception simulation models like Brettel's dichromat transformation model are available for simulating color perception for CVD patients, but are

primarily used for visualization and are not available for decision-making scenarios [7]. Various color-blind mobile applications and EnChroma glasses are available for providing limited aid, but are not available for integrating CVD detection, color detection of garments, and color perception-aware decision-making scenarios in a single framework.

In order to bridge this gap, the current research aims to develop a perception-aware fashion assistance system based on color vision deficiency type identification, garment color detection, and user perception modeling. In the first step, the system identifies the type of CVD of the user through a digital detection process. In the next step, computer vision is utilized to identify the dominant color of the garment based on images of clothing items. The identified color is then processed through a perception model based on machine learning and paired datasets of actual and perceived colors for different types of CVD. This perception model is trained to learn the relationship between actual and perceived colors for with CVD and provide a more accurate and confident fashion assistance system based on user perception of colors.

The study contributes to several of the United Nations' Sustainable Development Goals (SDGs). First, it contributes to SDG 10, which is "Reduced Inequalities." The study improves accessibility for people with CVD and facilitates equal participation for all in digital fashion platforms. The study also contributes to SDG 9, which is "Industry, Innovation, and Infrastructure." The study developed an innovative perception-aware system by integrating computer vision and machine learning technologies. Additionally, the study contributes to SDG 3, which is "Good Health and Well-being." The study improves the self-confidence and independence of people with limited visual perception.

The study also aligns with the Technology Integration & Management (TIM) research cluster. The study intersects with the Human-Computer Interaction, Usability Analytics, Requirements Engineering, and Ethical and Social Issues of Technology Integration domains. By integrating computer vision, machine learning, and perception-aware systems, the study is applied research aimed at improving accessibility, usability, and inclusivity in digital fashion recommendation systems.



Figure 1.1-color as seen by color blind users

1.2. Literature Review

There is considerable research in the areas of color vision deficiency detection, garment color recognition, perception simulation, and fashion recommendation systems. Nevertheless, most of the research is focused on resolving technical issues and not offering a comprehensive solution for people with color vision deficiency. In this section, we will discuss some of the research works in the areas of color vision deficiency, color detection, perception simulation, and technologies that help people with color vision deficiency.

A. Color Vision Deficiency and Its Impact

Color vision deficiency is another term for color blindness. Color vision deficiency is defined as "the inability to see colors." This is because people with color vision deficiency lack cone cells in their retinas. According to Birch, "the most common type of color vision deficiency is red-green color blindness, which affects 8% of men and 0.5% of women" [1]. This is often. Color vision deficiency affects many areas of life, for example, in selecting clothes.

People with color vision deficiency have difficulty recognizing the colors of clothing. This is easily recognizable for people with normal vision. Most digital fashion apps use the RGB color model. This is difficult for people with color vision deficiency. They often must seek help from people with normal vision. Although there is considerable research around color vision deficiency, most of it focuses on raising awareness and improving diagnosis.

B. Color Vision Testing and Type Identification

The Ishihara test is a technique for identifying color vision deficiency. This technique makes use of Ishihara plates, which are made up of colored dots that form numbers or other shapes. Only people with color vision are able to see these shapes clearly [3]. This technique can effectively identify color vision deficiency types such as protanopia and deuteranopia.

There are some studies that have attempted to digitize this technique. Tan and Zhang developed a system for color vision deficiency detection using digital Ishihara plates [4]. Kim and Park developed an app for color vision testing using smartphones [2]. Other researchers developed a technique for simulating color vision deficiency using image processing and deep learning [9].

Most of these techniques are for identifying color vision deficiency, but they are not smartphone-based color vision testing capable of solving daily life issues, such as choosing clothes. Therefore, it is still necessary to develop a technique for identifying color vision deficiency and for addressing fashion issues.

C. Automated Color Detection in Clothing Images

Accurate color detection is crucial for a perception- fashion system. Digital image processing techniques, such as color space transformations and clustering algorithms can efficiently identify the color in an image [5]. These algorithms can be used to classify colors in images [10].

For example, Wong et al. developed a system that can recognize the color of a shirt and display it as text. This can help people with color vision deficiency [6].

However, with this system, people with color vision deficiency may still misinterpret the color because the perceived color may differ from the actual RGB value.

D. Color Perception Simulation Models

Perception simulation is important for understanding how people with color vision deficiency see the world. Brettel et al. developed a simulation framework that can mimic the vision of people with color vision deficiency [7]. This framework uses the LMS color space and transforms the image's RGB values to reflect the perception of people with color vision deficiency.

However, most existing perception simulation models have been used for visualization and evaluation rather than for developing intelligent recommendation systems. Therefore, it is still necessary to develop a system that can use perception simulation results to provide fashion guidance for people with color vision deficiency.

E. Integrated Outfit Recommendation Systems for CVD Users

Pottekat et al. surveyed recommendation systems designed for color-blind users [8]. They found that effective systems must combine color compatibility rules, machine learning techniques and accessibility principles.

They also found that most existing systems only handle a component, such as color detection or compatibility. There is a need for a system that can integrate color vision deficiency type identification, color extraction, perception simulation and user feedback.

Such a system should include color vision deficiency type detection using digital Ishihara testing [4], automated dominant color extraction from clothing images [6], perception simulation for specific color vision deficiency types [7] and user-friendly color labels and compatibility scoring [8].

F. Existing. Services

There are tools available to help people with color vision deficiency. For example, the mobile app Color Blind Pal can help users recognize colors and EnChroma glasses can assist with color discrimination [14].

These tools are popular among users. Can provide basic help with color naming. However, they have limitations. They are primarily color-detection tools and lack clothing detection, personalization for different types of color vision deficiency, or perception-aware fashion options. Optical filters like EnChroma glasses are not effective for all users and do not provide outfit coordination or recommendation options.

Although much has been done in terms of color vision deficiency detection, clothing color recognition, and perception simulation, most of these studies are partial in nature, meaning they are not a complete package. There are a lot of studies that have focused on diagnosing CVD and recognizing colors, but none of them have tried to relate it to real-life situations like dressing up or receiving fashion advice. Currently, most of the assistive technologies for CVD patients are limited to basic color recognition and are not fully equipped to deal with the perception of colors for different types of CVD. Hence, there is a need for a comprehensive system that incorporates CVD type detection, clothing color detection, perception simulation, and fashion advice for people with color vision deficiency.

Although much research has been done on the detection of color vision deficiency, clothing color recognition, and perception simulation, the majority of the work has been done on individual components. Thus, there is a need to propose a unified system for identifying color vision deficiency types, detecting clothing colors, and considering perception in color interpretation to enhance decision-making in fashion for color vision deficiency patients..

1.3. Research Gap

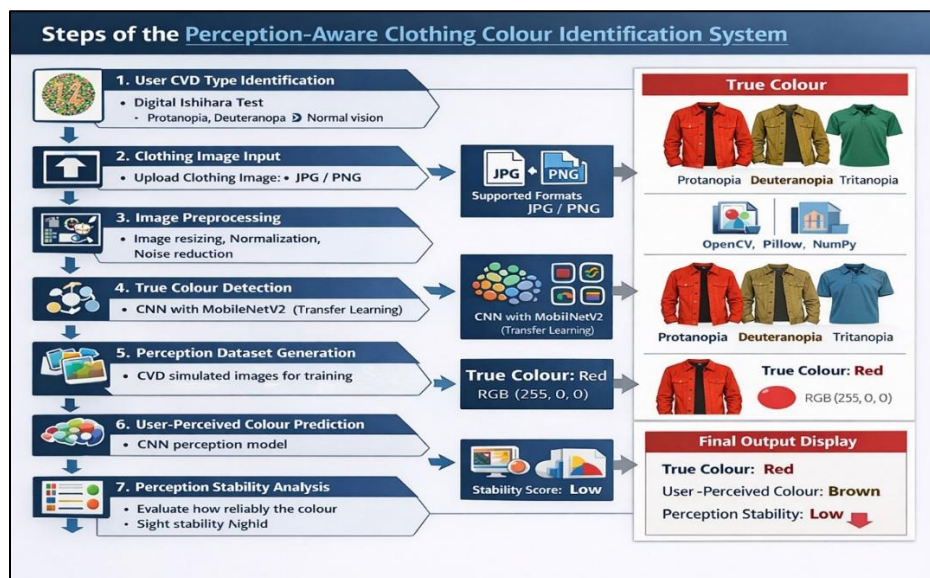
Although many research works have been conducted to explore Color Vision Deficiency (CVD) detection, clothing color extraction, and color perception simulation, these functions are usually considered individually rather than. For instance, clinical tools like the Ishihara test can effectively identify various types of Color Vision Deficiencies [2]-[4]. Nevertheless, such tools are mostly utilized in clinical diagnosis and are not usually considered in digital fashion applications. Similarly, image processing techniques can identify the dominant color information of garments using color space transformations and clustering [5], [6]. Nevertheless, most of these techniques focus on extracting a limited set of dominant colors and ignore important factors such as background pixels, lighting conditions, and shadow effects, which may not accurately reflect the colors perceived by the user. Similarly, LMS-based simulation models [7] can approximate how individuals with CVD perceive colors, but they are used for visualization purposes rather than integrated into structured fashion guidance systems. Existing fashion recommendation platforms focus on aesthetic matching and general color harmony [8], [10], without identifying the user's specific CVD type or translating garment colors into clear "True Color vs. You See" representations. Enhancement-based approaches, including deep learning-based color modification methods [9], aim to improve visibility; however, they may introduce visual distortions, reduce interpretability, and require substantial computational resources, making them less suitable for real-time fashion platforms.

Besides, most color detection systems provide objective color values in terms of RGB and HEX. These offer interpretational difficulties for people with CVD. LMS perception simulation models can offer an approximation of how people with CVD perceive colors [7]. These LMS simulation models have mostly been used for visualization rather than as part of actual fashion assistance systems. Most of the existing fashion recommendation systems have focused on aesthetics and overall color harmony [8], [10]. They do not identify the type of CVD and do not offer an interpretation of colors for perceptions.

Due to the fragmented nature of existing approaches, existing solutions have not been effective in facilitating clothing selection for people with CVD. Although existing solutions have shown promise for CVD classification [2]–[4] and clothing color detection [5], [6], people with CVD may still face difficulties in determining color compatibility from their perspective of color perception.

Therefore, there is a research gap in designing an integrated perception-aware fashion assistance system that incorporates automated clothing color detection, user-specific CVD type identification, and color perception simulation into one framework. This research aims to fill in the research gap by integrating computer vision techniques using OpenCV for image preprocessing, deep learning techniques using TensorFlow/Keras with MobileNetV2 transfer learning for clothing color detection, and Color spacious for physiologically accurate CVD perception simulation. By providing users with structured labels such as “True Colour” and “User-Perceived Color,” the proposed system provides users with clear and interpretable color information that can help people with CVD in making confident decisions.

Figure1. 2-step-by-step flow of the component



Evaluation Criteria	CVD-GAN (Research Model)	Daltonize (Tool)	Colorblind Palette Generator	Existing Fashion Apps	Proposed Perception-Aware System
Automated CVD Type Detection	✗	✗	✗	✗	✓
Garment-Level Color Detection	✗	✗	✗	✓	✓
Fashion-Specific Simulation	✗	✓	✗	✗	✓
Personalized Perception Modelling	✓	✓	✗	✗	✓
Structured True → Perceived Color Mapping	✗	✗	✗	✗	✓
Recommendation Integration	✗	✗	✗	✓	✓

Table 1-comparison between existing systems

2. OBJECTIVE

2.1 Main Objective

To develop a perception-aware system that detects clothing colors and predicts how they appear to individuals with Color Vision Deficiency.

The major aim of this research is to create a perception-aware system that can help Color Vision Deficient (CVD) individuals understand how they see the color of clothing they are wearing. The system, using deep learning and computer vision concepts, can identify the actual color of clothing and then forecast how it is perceived by individuals suffering from Color Vision Deficiency of a specific kind. The system can also provide collated output and check the stability of the perception of those colors.

2.2 Specific Objectives

1. Identifying the user's Colour Vision Deficiency (CVD) type

This step is intended to identify the Color Vision Deficiency type of the user. To do this, the digital version of the Ishihara color vision test is carried out. This is done by displaying the user the Ishihara plates via the web interface and then recording their responses to ascertain whether the user has normal color vision or has any of the following Color Vision Deficiency types: Protanopia, Deuteranopia, or Tritanopia. Once this is done, the user can then use this to forecast how he or she would view the color. .

2. Preprocess clothing images

The goal of this step is to pre-process the clothing images uploaded for color detection. This is done by applying conventional image processing techniques using computer vision libraries such as OpenCV, Pillow, and NumPy. This helps to improve the quality of the image, hence improving the performance of the machine learning algorithm for color detection.

3. Detect the actual color of the clothing using deep learning

The main aim of this objective is to detect the actual color of the clothing in the image of clothing that is uploaded. For this purpose, a Convolutional Neural Network (CNN) is utilized with TensorFlow and Keras. With this, it is possible to take advantage of transfer learning and utilize MobileNetV2 to extract visual features from images of clothing. At last, the actual color of the clothing is determined and classified as red, blue, green, or black.

4. Prediction of perceived color for diverse CVD types

For this purpose, it is required to perform a prediction of the color perceived by people suffering from diverse types of color vision deficiency. For this purpose, a perception-aware dataset is created, followed by simulating color vision deficiency effects on the dataset of the image of clothing using Colorspacious. With this, it is possible to obtain an image based on the perception

of color under diverse types of color vision deficiency. For this purpose, a convolutional neural network is utilized, namely MobileNetV2. After this, it will be able to take as input an image of the clothing and the corresponding color vision deficiency type and proceed to provide the color perception.

5. Generate perception-aware color labels and perception stability levels

This is an objective of the system: to generate structured color labels and compare them with the true color, and to effectively communicate color information to users in a manner that is easily understood. The system can be able to generate structured color labels and compare them with the true color and perceived color. The output is presented in a specific format, for example: True Color → User-Perceived Color

The system can also be able to generate perception stability levels based on the difference between true and perceived colors. The system can classify the level of color stability as High, Medium, or Low.

3. METHODOLOGY

Research Approach

This research utilizes the Design Science Research (DSR) method to design, test, and validate the development of the perception-aware clothing color identification system for individuals with Color Vision Deficiency (CVD). The aim of the research is to produce an applicable artefact capable of utilizing deep learning technology for clothing color detection, perception-aware color prediction, and color labeling, with clear computational models. The developed system utilizes computer vision models and the CNN-based MobileNetV2 transfer learning model to identify the true color of the garment and the color as perceived by the individual, based on the CVD type. An experiment has been conducted to assess the color detection accuracy, color prediction, and usability of the developed system.

Methods and Algorithms

1. CVD Type Identification

The user's color vision capability is verified with the aid of a digital Ishihara chart. The user has to identify the numbers he or she can see, and the scoring process is simple and rule-based. Depending on the results, the user falls into one of the four categories: protanopia, deuteranopia, tritanopia, or normal vision. The simple, rule-based approach has been adopted because it reflects the real process.

2. Garment Color Detection

uploaded the image of the clothes, we use image preprocessing using OpenCV. This helps to resize the image to a standard form, which can be processed by the model. This enhances the quality of the image and makes it consistent across all inputs.

The next step is to identify the color of the clothes using a Convolutional Neural Network, developed using TensorFlow and Keras. MobileNetV2 is used, and transfer learning is employed to extract meaningful features from the image of clothes. This reduces the need to train the model extensively, thereby reducing the amount of data required. Once the image has been processed, the model identifies the color of the clothes by assigning a color label to the RGB values.

3. User Perception Color Prediction

This section is concerned with how the appearance of clothes is presented to the user with various Color Vision Deficiencies. To start with, a perception-aware dataset is created with the Colorspacious library, which creates the appearance of colors of the garment under various CVD conditions such as protanopia, deuteranopia, and tritanopia. These images will be representative of the appearance of the color to the color-blind user.

A deep learning model will be trained on the dataset with the aim of predicting the colors of the garment with the appearance aspect considered. A Convolutional Neural Network, implemented with TensorFlow and Keras, utilizing the MobileNetV2 transfer learning method, will be used to extract the features of the garment and the appearance of the color with reference to the CVD type.

At runtime, the model will be fed with the image of the garment and the specific type of CVD, and the predicted label will be generated on the appearance of the color of the garment.

Tools and Platforms

1. Programming Environment

Python is adopted as the development language because it has an extensive collection of computer vision and machine learning libraries.

2. Image Processing and Segmentation

For image processing, garment images are processed using OpenCV (OpenCV-python), which is used to carry out image preprocessing activities.

For image segmentation, it is ensured that the images are appropriate for deep learning analysis. Image loading and saving operations are handled using Pillow (PIL) since it allows various image formats such as JPG and PNG, while NumPy is used for efficient numerical computation and pixel array manipulation during processing.

3. Dominant Color Extraction

In order to provide a flexible environment, deep learning module is initiated using TensorFlow and Keras so building and training neural network can be done efficiently. A CNN architecture is used for recognize images and MobileNetV2 as a pretrained model for transfer learning. It provides feature extraction with lower cost while maintaining high classification accuracy.

4. Color Perception Simulation

Color perception simulation is implemented using the ColourSpacious library, which supports LMS-based color vision models. It allows accurate stimulation of colors perceived by different types of CVD individuals.

5. Data processing and visualization

To split the data set into training and testing sets for train and evaluate the model, Scikit-learn is used. Matplotlib is used to visualize accurate model performance, including training accuracy and loss curves to analyze the results of color detection and perception stimulation.

Data collection and Ethical consideration

Preliminary data will be collected before implementing the system to efficiently address the user need, along with to validate the problem which affects colour blind individuals. The target population will be selected through social media platforms, colour blind communities and mainly through coordination with the Eye clinic, National Hospital of Sri Lanka. Here, preliminary data includes the patient's basic demographic data (Age and Gender), vision status, challenges faced when choosing clothing and the need for a perception-aware system. This data will be collected using a structured Google form or a physical form. All data will be processed in accordance with ethical principles. Before data collection, participants will be made aware of the purpose of the research and will be asked for their consent prior to participation. Collected personal data will be used only for research purposes and will be treated confidentially.

After implementing the system test will be carried out to collect data further to measure the accuracy of the system. These tests include correct identification of CVD type; how similar the stimulated image represents the colours that are seen by CVD individuals, and the comparison between true colour and perceived colours. Quantitative analysis will be done using CVD classification accuracy percentage, mean perception similarity score, and perceived usability satisfaction average. Through open-ended feedback, qualitative data will be assessed.

After recording data in a data collection sheet, a specific Microsoft Excel sheet is used to record data that can be upload onto SPSS for further analyzation. Depending on the p value parametric and non-parametric tests can be run to consider the study as statistically significant.

Validation strategy and performance metrics

The system will be evaluated through both technical validation and user-based evaluation. Technical validation focuses on CVD classification accuracy, clothing color detection accuracy, and consistency of perception stimulation. User-based validation will be tested with 20-30 participants. They will use the system and rate how similar the simulated image is to how they see colors, using a Likert scale. The system will be evaluated based on: CVD classification accuracy percentage, Mean perception similarity score, Color detection consistency percentage, and average usability satisfaction score. In addition, participants' feedback will be collected to identify the strengths and weaknesses of the system and improve it accordingly. It is ensured that the system is tested from a technical as well as a user's perspective.



Figure 3.1-Google Form Google form link: <https://forms.gle/uKK3ERgXHLvqJUc3A>

4. HIGH-LEVEL SYSTEM ARCHITECTURE

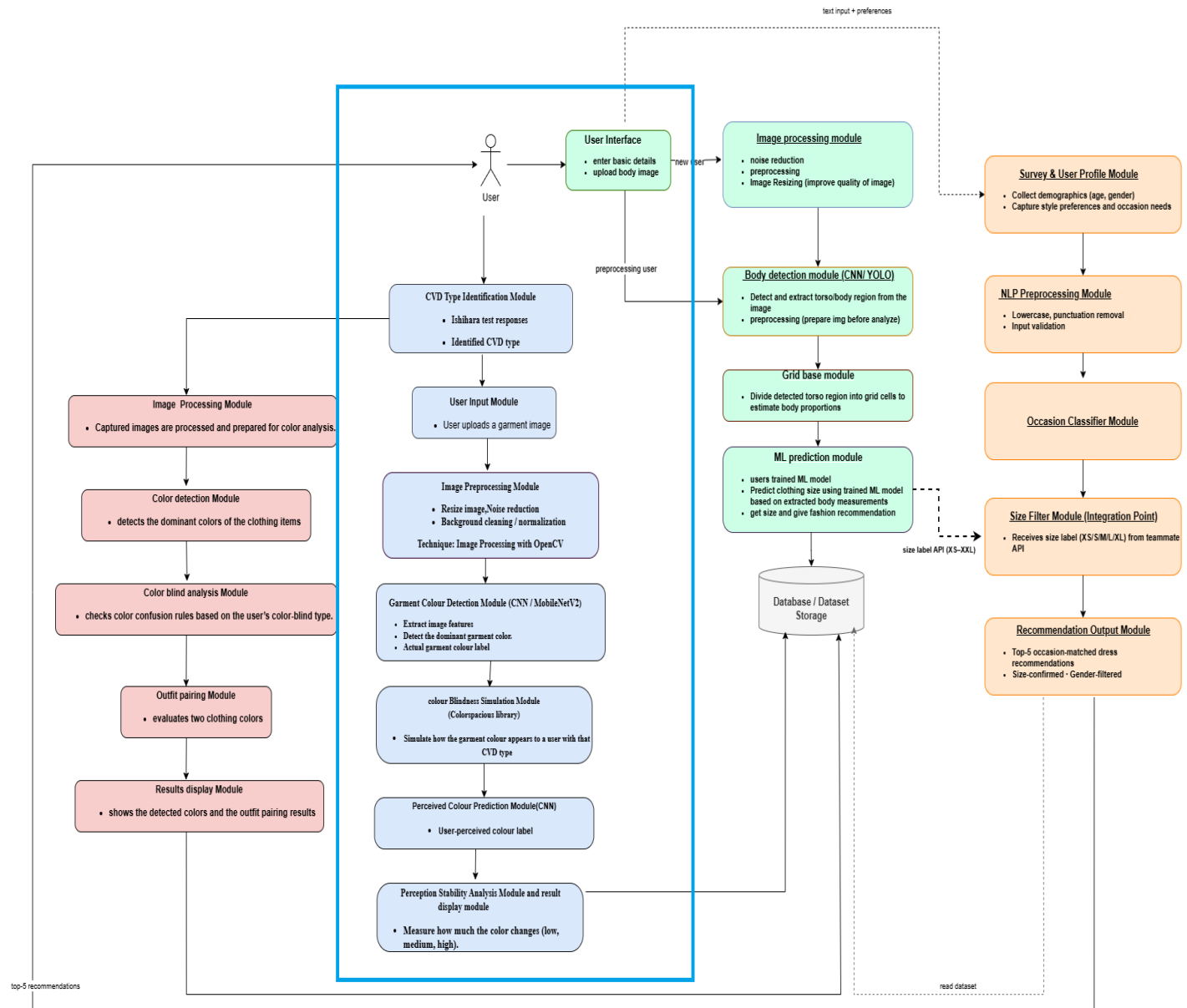


Figure 4.1-Project Overall System Diagram

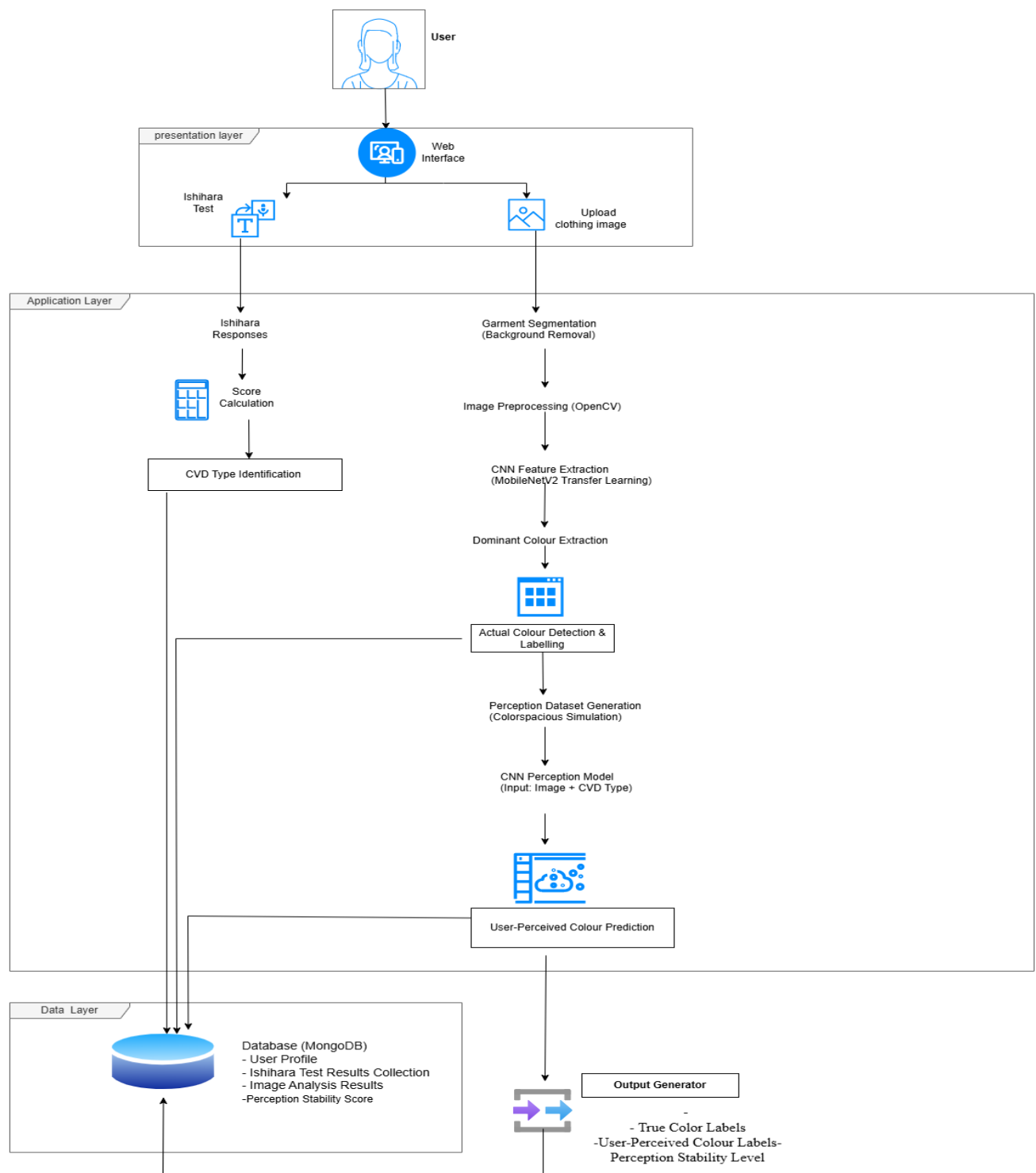


Figure 4.2-actual and user-perceived color recognition based on CVD

The system starts by detecting your Color Vision Deficiency type through a digital version of the Ishihara test presented on the interface. The responses are then used to determine whether your color vision is normal or whether you are suffering from Protanopia, Deuteranopia, or Tritanopia through a rule-based scoring system. After detecting your color vision deficiency type, you are required to upload an image of your clothing, which is then used as input to analyze your color vision deficiency.

The image is then subjected to preprocessing using OpenCV, Pillow, and NumPy to prepare it for deep learning techniques. After preprocessing your image, a Convolutional Neural Network (CNN) is used to analyze your image and identify your actual color. This is achieved through TensorFlow/Keras using MobileNetV2 transfer learning to identify the key features from your image and determine your actual garment color.

To determine how your color is perceived by you given your color vision deficiency type, a perception-aware dataset is used to simulate how you would perceive your color given your color vision deficiency type. This is achieved through a color vision simulation library called Colorspacious. Using this dataset, your actual color is determined based on image and color vision deficiency type. The perception stability is then determined by comparing your actual color with perceived color. The stability is then classified as high, medium, or low based on how different your actual color is from your perceived color. The result is then presented to you in a user-friendly format to help you understand your color vision deficiency from your own perspective.

5. USER REQUIREMENTS

5.1. Stakeholder Identification

The primary stakeholders. The main stakeholders of this system would be people with Color Vision Deficiency. They are protanopia, deuteranopia, and tritanopia. People with Color Vision Deficiency cannot tell different colors apart. They get confused when they see different colors. This system would be helpful to people with Color Vision Deficiency, as it would help them identify the color of clothing and how it would appear to them, depending on the Color Vision Deficiency they have. It would also help them know whether it is safe to wear certain clothes depending on how colors appear to them. This system would be very helpful to people with Color Vision Deficiency.

The secondary stakeholders of the proposed system would be people who would like to use it for their fashion needs. People without Color Vision Deficiency would be affected by the proposed system. This is because it would help them understand the differences in how people perceive colors. The proposed system would be of great significance to people who want to shop online. This is because it would help them to avoid any misunderstanding that may arise from the color of the products they purchase.

Institutional stakeholders may include eye care institutions, such as eye hospitals across the country, or color-blind communities. They would help provide validation support, recruit participants, conduct user testing, and gather feedback, thereby enhancing the reliability and usability of the system.

5.2. Functional requirements

FR1 – CVD Type Identification-The system shall be able to identify the type of Color Vision Deficiency of the user through a digital version of an Ishihara test based on colors. It shall be able to classify the user as having normal vision or any of the Color Vision Deficiencies: Protanopia, Deuteranopia, or Tritanopia.

FR2 – Clothing Image Acquisition-The system shall be able to acquire clothing images from users through an image upload facility from image formats such as JPG or PNG.

FR3 – Image Preprocessing: The system shall preprocess images using OpenCV and NumPy libraries for resizing, normalization, and standardization.

FR4 – True Garment Color Detection

The system shall be able to identify the true color of the garment from an image of clothing using a Convolutional Neural Network.

FR5 – Color Label Generation- The system transforms the color value into simple and easily understandable terms, such as red and blue, which can be easily interpreted and understood by human beings.

FR6 – Perception-Aware Color Prediction

The system will be able to predict how the garment's color will be perceived by individuals with color vision deficiencies using a deep learning model and a perception-aware dataset generated through color vision simulation.

FR7 – Perception Stability Analysis

The system will determine how different the actual color is from the color we think we see. Then it will tell us how reliable that perception is, by saying it is Medium or Low stability.

5.3. Non-Functional requirements

1. Usability

The system should be simple and user-friendly so that anyone with Color Vision Deficiency is able to use it without any confusion. It should be very easy to use so that even a person with Color Vision Deficiency is able to use it without any difficulty. It is important that colors and colors be easily understandable.

2. Performance

Color detection from clothing images should be fast so that it gives results within 3 to 5 seconds. It should be able to work on normal computers without any difficulty or slow performance.

3. Accuracy

It should be able to accurately identify clothing colors through a computer model using a convolutional neural network with MobileNetV2 transfer learning. It should be able to accurately represent colors to a person with Color Vision Deficiency.

4. Security and Privacy

It should be able to protect user data and only use it when it is needed. It should not use any other records apart from what is needed. It should be able to store only Color Vision Deficiency results and other data from users with consent to ensure proper functionality.

5. Compatibility

It should be able to work on Google Chrome or Microsoft Edge web browsers and on desktop computers so that it is easily accessible to anyone.

6. COMMERCIALIZATION PLAN

The plan is to develop a perception-aware fashion recommendation system that is to be offered as a SaaS solution to those who suffer from Color Vision Deficiency (CVD) and fashion businesses. The target group would include color-blind persons who find it difficult to differentiate the colors of clothes and would require assistance in shopping on their own, especially when shopping online. The second target group would include e-commerce businesses, online fashion businesses, fashion brands, and accessibility businesses that seek to improve user experience and reduce product return rates due to color-related confusion.

The system closes the gap that exists between real colors and perceived colors. Based on this, it can help identify the user's type of color vision deficiency (CVD), thus modeling how they perceive clothes of various colors. It, therefore, offers a two-color system that is perception-based in nature. This makes a user confident and independent, especially when it comes to color-blind people. For businesses, it can help in increasing satisfaction and reducing purchase errors.

The revenue strategy is based on a hybrid approach. The freemium strategy is followed by offering basic services like CVD recognition and clothing color detection for free. This is to enhance user engagement. The premium services like user perceived color prediction and perception stability analysis will be offered for a price ranging from LKR 1,000 to LKR 1,500 per month. In addition to that, a B2B licensing option is also provided to fashion retailers and e-commerce websites to use the perception-aware color module in their system.

The estimated cost of commercialization includes software development, cloud hosting, database management, API maintenance, and cybersecurity. The operational cost of running the system is estimated at LKR 15,000-30,000 per month, depending on usage. Other costs may include marketing through social media and working with organizations for accessibility. Perception-awareness is the key competitive advantage of the proposed system. CVD identification, clothing color identification using deep learning techniques, and user perception of color prediction are combined into a single platform. The innovation is represented in two colors: True Color and User-Perceived Color. This innovation can be protected under copyright and trademark laws.

7. BUDGET AND JUSTIFICATION

The cost of the project is estimated to be LKR 148,000. This cost covers all the basic expenses of a system to successfully complete the project. The use of cloud services ensures system availability, which helps in successfully conducting tests. The cost of hardware is essential for safe storage of data and for a smooth process of computation, such as segmentation of images and simulation of colors. The cost of software is essential for safe access to a domain and for proper support in evaluation. The cost of human validation is essential for data collection from participants and for proper management of the process by the hosting institution.

Description	Cost (LKR)
Cloud Services	
Cloud Hosting & Deployment (10 months × 6,000)	60,000
Hardware Costs	
External Hard Drive / Backup Storage	15,000
Development Equipment Support (System upgrades/peripherals)	12,000
Software Licenses / Tools	
Domain & SSL Certificate	8,000
Testing & Evaluation Tools (Optional APIs / Services)	10,000
Human Effort & Validation Cost	
Participant Data Collection & Survey Support	20,000
Travel & Institutional Coordination	15,000
Documentation & Miscellaneous	
Documentation & Printing	8,000
Contingency for unexpected expenses	10,000
Total Estimated Budget	148,000

8. WORK BREAKDOWN STRUCTURE (WBS)

The project is structured using a hierarchical Work Breakdown Structure (WBS) to ensure systematic planning, balanced workload distribution, and alignment with the 11-month project timeline. The WBS divides the research into clearly defined phases, beginning with project initiation and progressing through design, implementation, testing, validation, and final delivery.

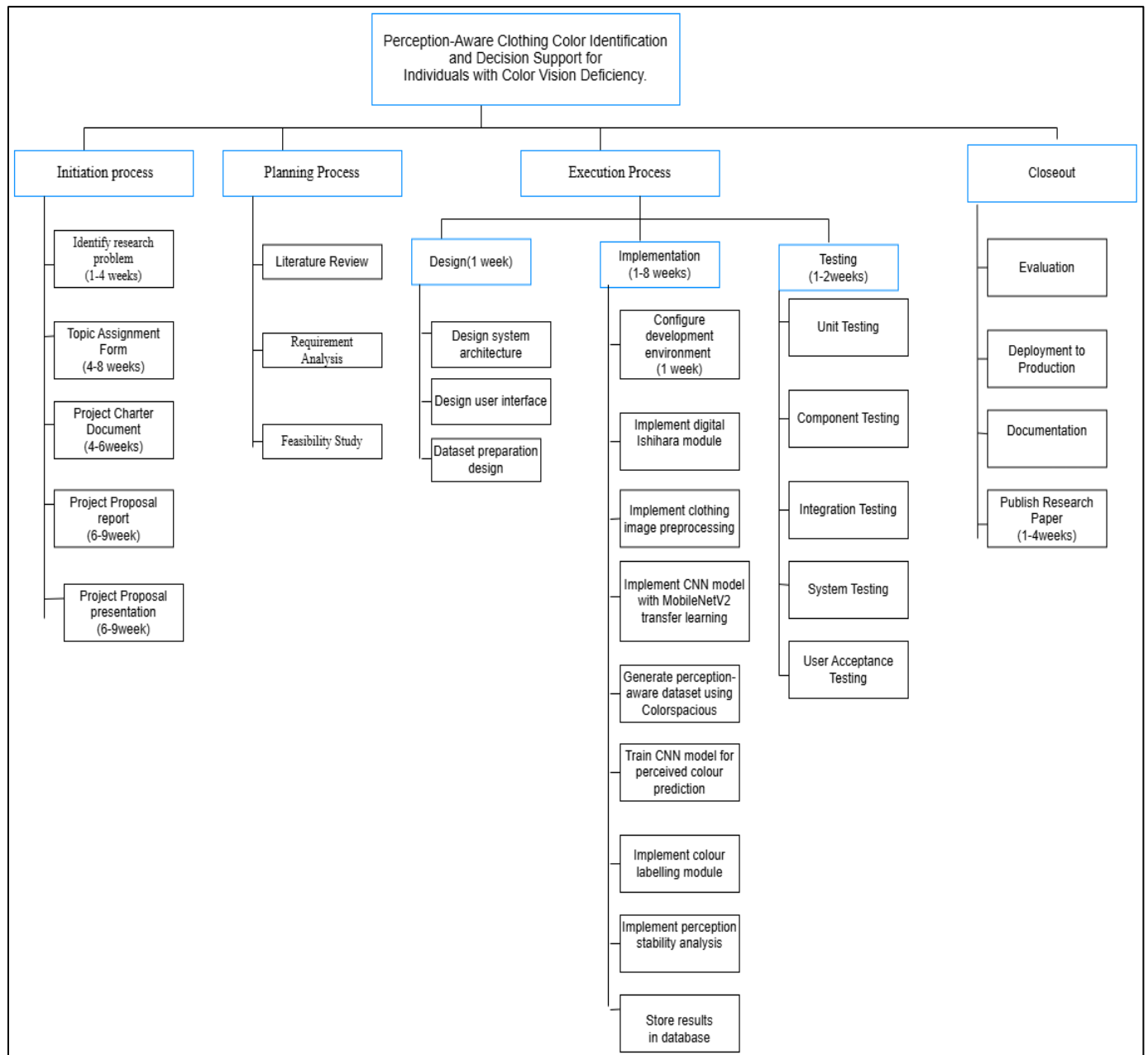


Figure 8.1-Work breakdown Chart

9. GANTT CHART

We hope to complete our research within this time limit.

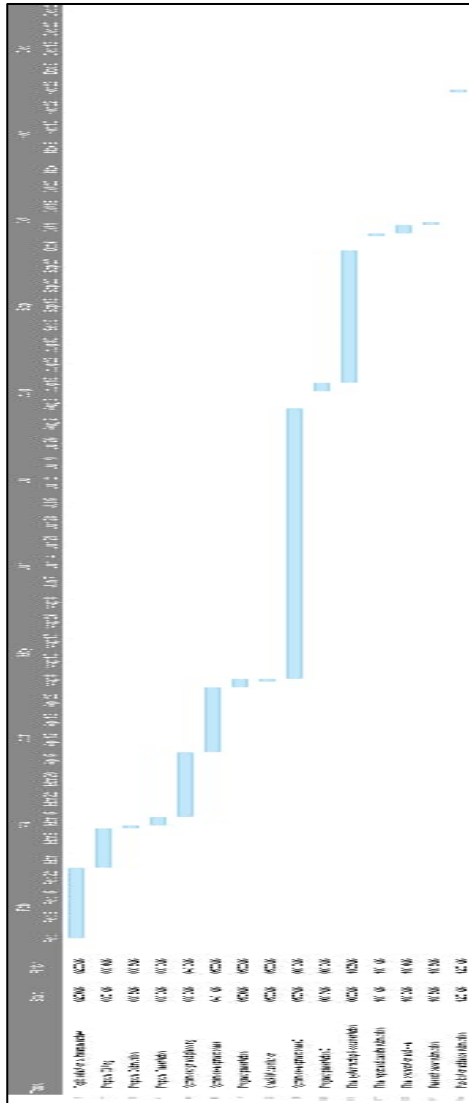


Figure 9.1-Gantt Chart

Gant chart Link:<https://app.smartsheet.com/sheets/RHRCmQxRcQVFrGwPXvRHVh2j9Q3MMX6QgWHMg561>

Google Drive Link:

<https://drive.google.com/file/d/1XHbuUjWiHrdB5wmzZzKGW6jRIZWwActR/view?usp=sharing>

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11.APPENDICES

11.1.Survey Instrument

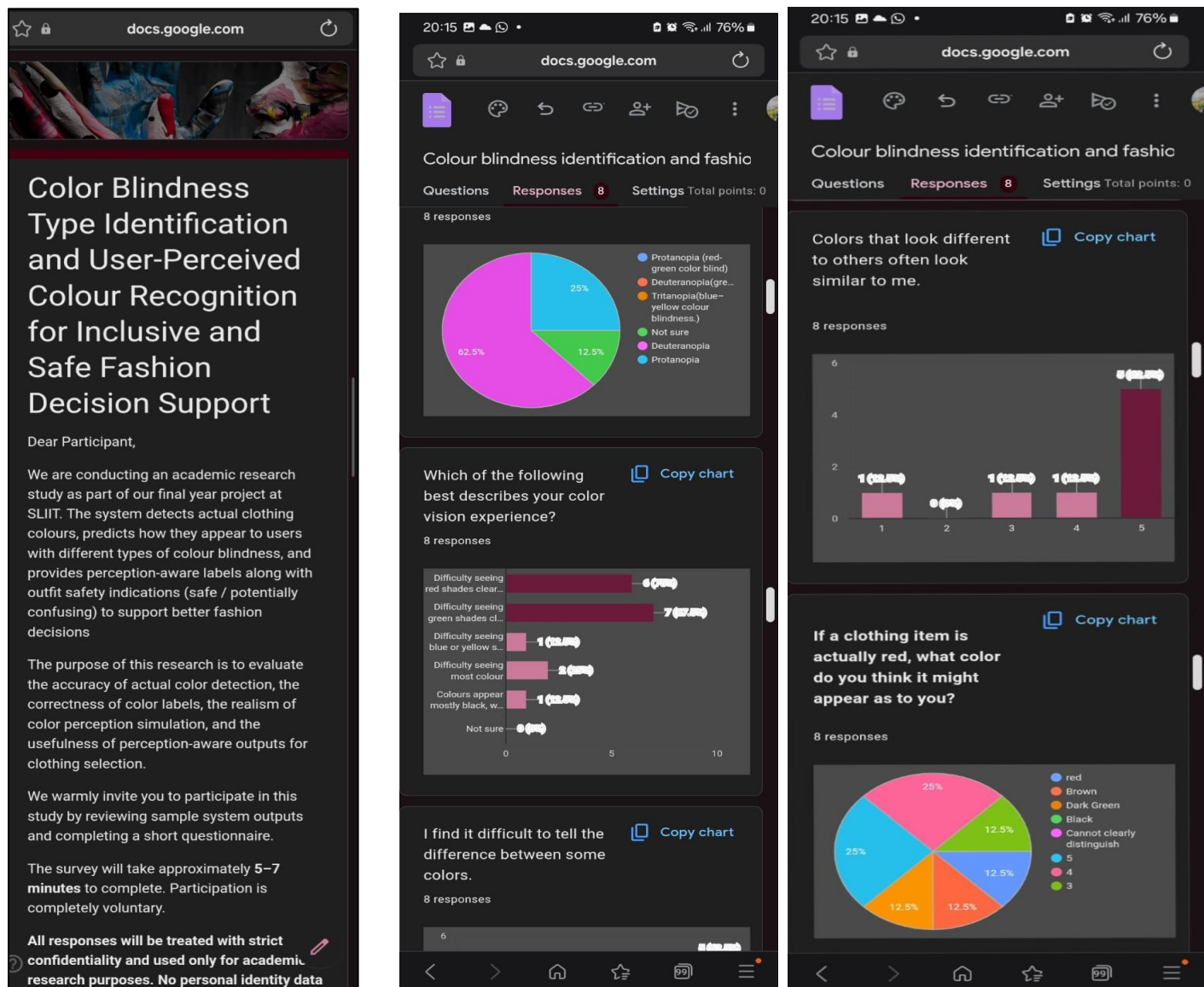


Figure 11.1-Google form and response

Google Form : <https://forms.gle/iqb2tc5HuMfFDPV98>

11.2. Detailed diagrams

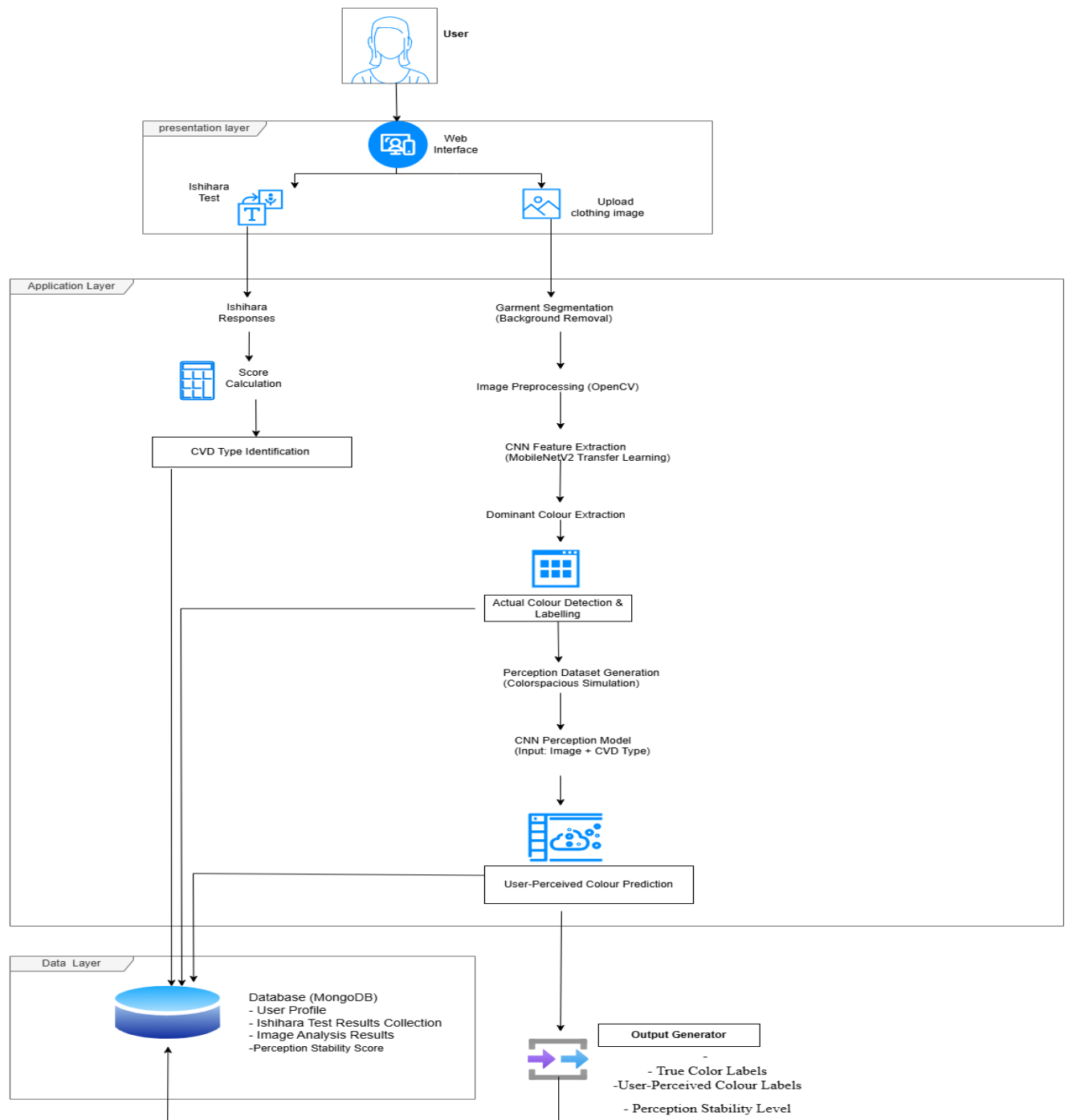
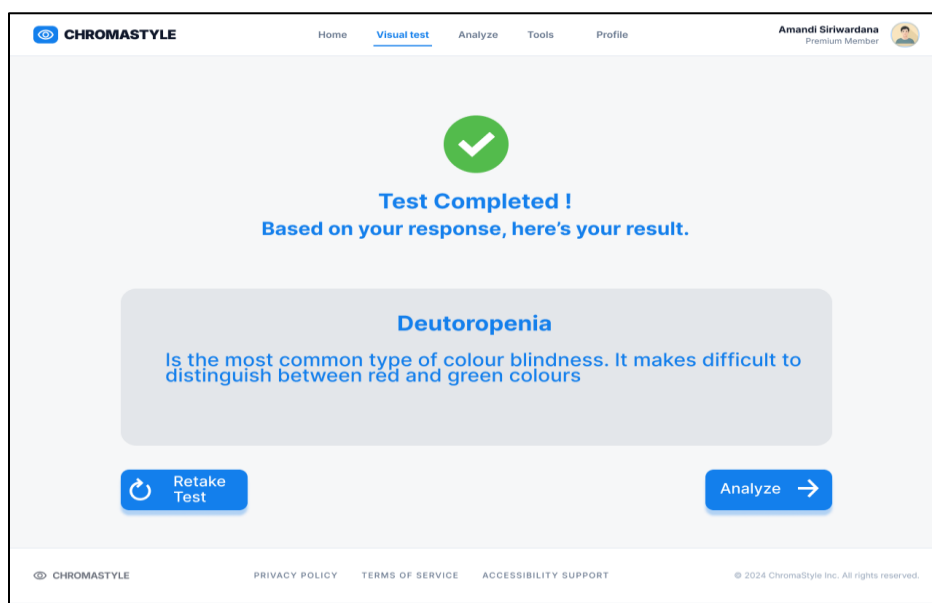
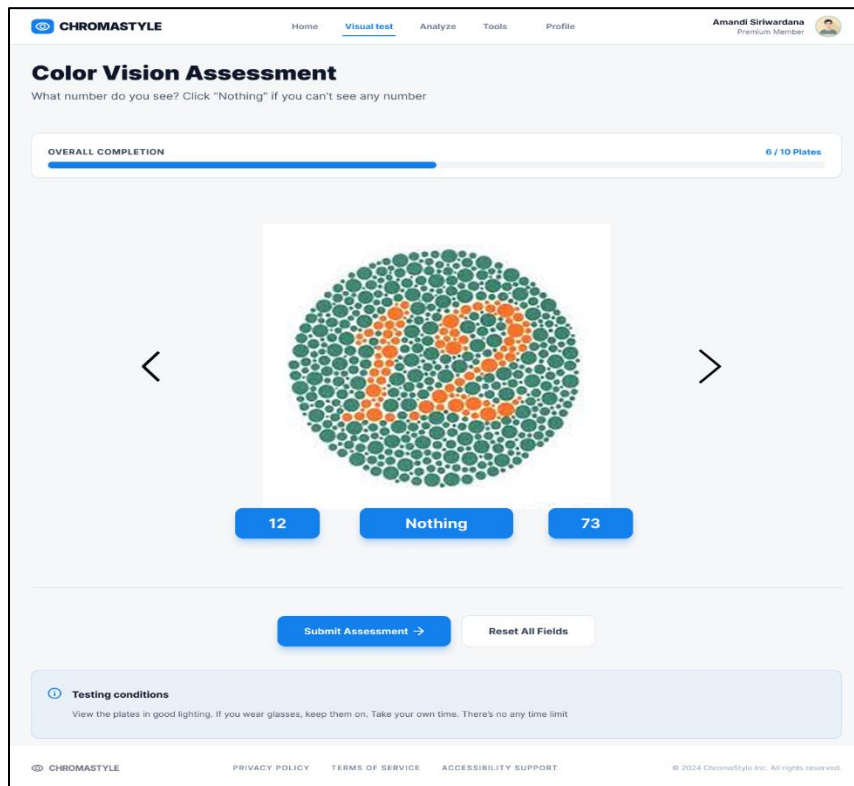
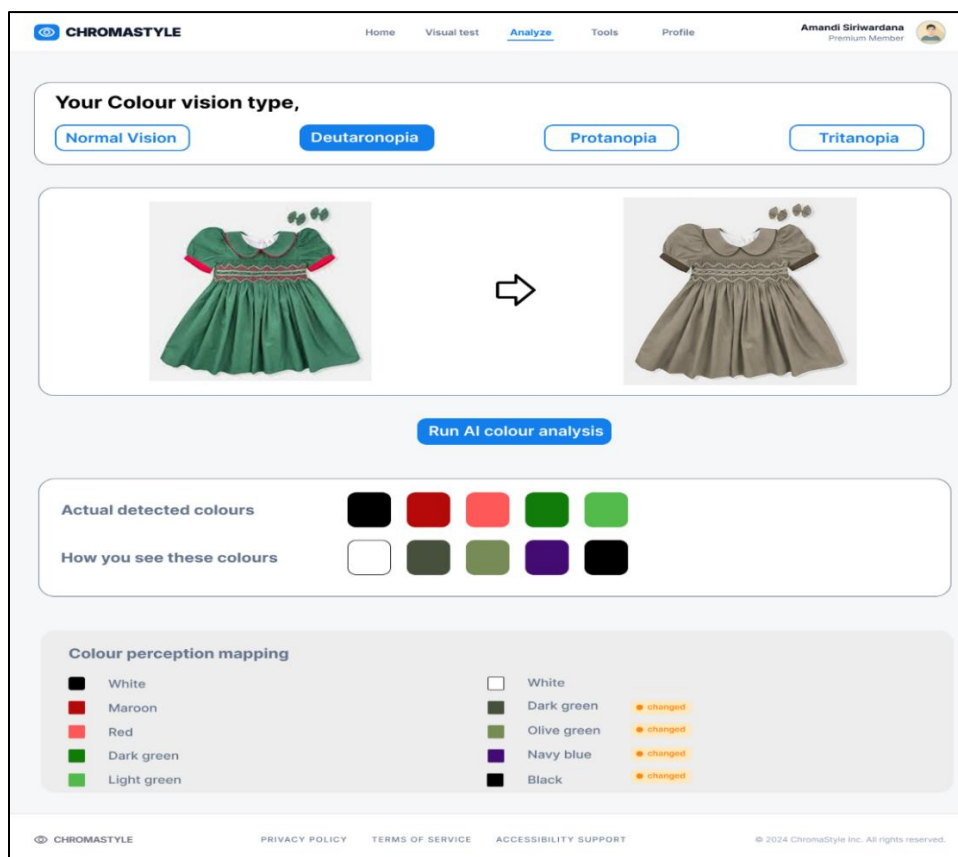
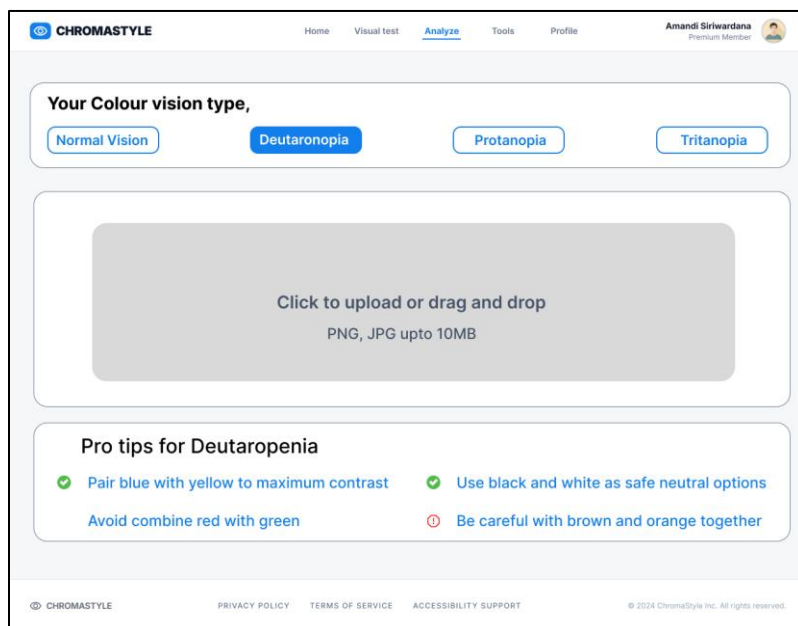


Figure 11.2-actual and user-perceived color detection based on CVD type

11.3. Preliminary UI mockups

Figure 51.3-Figma UI Designs





11.4. Data gathering

13/03/2026

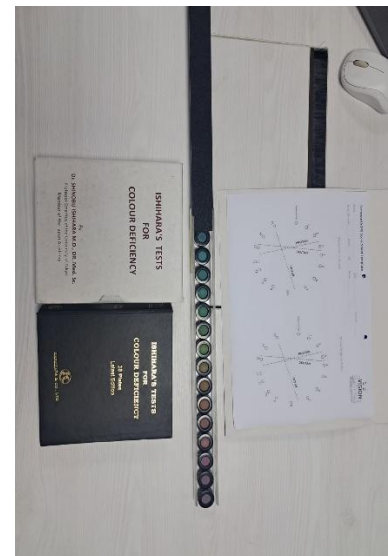
Information Confirmation Letter.

I hereby confirm that I have provided information and shared my experiences regarding colour vision deficiency for the research study conducted by students of SLITT. (IT22242372, IT22098696).

I understand that the information I shared will be used only for academic research purposes. I voluntarily participated in this discussion to support the study.

Name: Vidya Jayarathna, Head of optometry.
Signature: 
13/03/2026.

Figure 61.4-Information Gather letter



CVD Identify tools

